

Problem Set 2

Applied Stats II

Due: February 18, 2026

Instructions

- Please show your work! Include the R code used for calculations.
- Submit your homework in .pdf form on GitHub.
- The assignment is due before 23:59 on Wednesday, February 18, 2026. Late submissions will not be accepted.

We are interested in what types of international environmental agreements people support (Bechtel and Scheve 2013). The dataset `climateSupport.RData` contains 8,500 observations on whether individuals support a policy, varying (1) the number of countries participating and (2) sanctions for non-compliance.

- Response variable:
 - `choice`: 1 if the individual agreed with the policy; 0 if not.
- Explanatory variables:
 - `countries`: Number of participating countries [20, 80, 160 of 192].
 - `sanctions`: Sanctions for missing emission reduction targets [None, 5%, 15%, 20%].

1. Additive Logistic Regression Model

- (a) Fit the additive model, provide summary output, global null hypothesis, and p -values.

```
1 load(url("https://github.com/ASDS-TCD/StatsII_2026/blob/main/datasets/
  /climateSupport.RData?raw=true"))
2 head(climateSupport)
3 str(climateSupport)
4 climateSupport$choice_bin <- ifelse(climateSupport$choice == "
  Supported", 1, 0)
5 model <- glm(choice_bin ~ countries + sanctions,
6               data = climateSupport,
7               family = binomial)
8 summary(model)
9 anova(model, test = "Chisq")
```

The number of countries is a strong predictor of policy support. The Chi-square test indicates a significant effect ($p < 0.001$), so we reject the null hypothesis that the number of countries has no effect. The level of sanctions is also significant (Chi-square = 68.426, $p < 0.001$), indicating that sanctions affect policy support. The null hypothesis that sanctions have no effect is rejected.

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL				8499	11783
countries	2	146.724		8497	11637 < 2.2e-16 ***
sanctions	3	68.426		8494	11568 9.272e-15 ***

Coefficients:

	Estimate	Std. Error	z	value	Pr(> z)
(Intercept)	0.07545	0.02537	2.975	0.00293	**
countries.L	0.45325	0.04373	10.364	< 2e-16	***
countries.Q	-0.05758	0.04419	-1.303	0.19258	
sanctions.L	-0.13745	0.04396	-3.126	0.00177	**
sanctions.Q	0.18790	0.04395	4.275	1.91e-05	***

2. Interpretation of Significant Variables

- (a) For 160 countries, how does increasing sanctions from 5% to 15% affect odds of support?

```
1 model <- glm(choice ~ countries + sanctions,
2               data = climateSupport,
3               family = binomial)
4 summary(model)
5 newdata_5 <- data.frame(
6   countries = factor("160 of 192", levels = levels(climateSupport$
  countries)),
```

```

7   sanctions = factor("5%", levels = levels(climateSupport$sanctions))
8 )
9
10  newdata_15 <- data.frame(
11    countries = factor("160 of 192", levels = levels(climateSupport$
12      countries)),
13    sanctions = factor("15%", levels = levels(climateSupport$sanctions)
14  )
15  log_odds_5 <- predict(model, newdata_5, type = "link")
16  log_odds_15 <- predict(model, newdata_15, type = "link")
17  odds_ratio <- exp(log_odds_15 - log_odds_5)
18  odds_ratio

```

Increasing sanctions from 5% to 15% reduces the odds of supporting the policy by about 28%, holding the number of countries constant. This matches the model output and shows the negative effect of higher sanctions.

- (b) For 20 countries, how does increasing sanctions from 5% to 15% affect odds of support?

```

1  newdata_5_b <- data.frame(
2    countries = factor("20 of 192", levels = levels(climateSupport$
3      countries)),
4    sanctions = factor("5%", levels = levels(climateSupport$sanctions))
5  )
6  newdata_15_b <- data.frame(
7    countries = factor("20 of 192", levels = levels(climateSupport$
8      countries)),
9    sanctions = factor("15%", levels = levels(climateSupport$sanctions)
10  )
11  log_odds_5_b <- predict(model, newdata_5_b, type = "link")
12  log_odds_15_b <- predict(model, newdata_15_b, type = "link")
13  odds_ratio_b <- exp(log_odds_15_b - log_odds_5_b)
14  odds_ratio_b
15  log_odds_5_b <- predict(model, newdata_5_b, type = "link")
16  log_odds_15_b <- predict(model, newdata_15_b, type = "link")
17  odds_ratio_b <- exp(log_odds_15_b - log_odds_5_b)
18  odds_ratio_b

```

For 20 participating countries, increasing sanctions from 5% to 15% also reduces the odds of supporting the policy by about 28%, holding the number of countries constant. The effect is similar because the interaction is not significant.

- (c) Estimated probability of support if 80 countries participate with no sanctions?

```

1  newdata <- data.frame(
2    countries = "80 of 192",
3    sanctions = "None"
4  )
5  predicted_prob <- predict(model, newdata, type = "response")

```

```
6 predicted_prob
```

Given 80 countries participate and no sanctions, the predicted probability of support is approximately 52.5%.

3. Interaction Term Between Countries and Sanctions

- Test whether including an interaction term is appropriate.

```
1 model_additive <- glm(choice ~ countries + sanctions ,  
2                       data = climateSupport , family = binomial)  
3 model_interaction <- glm(choice ~ countries * sanctions ,  
4                          data = climateSupport , family = binomial)  
5 anova(model_additive , model_interaction , test = "Chisq")
```

The likelihood ratio test shows that the interaction between countries and sanctions is not significant ($p = 0.786$). This indicates that the effect of sanctions on policy support is roughly the same at all levels of country participation. Including the interaction does not improve the model, so the answers to 2a and 2b remain unchanged.

```
Model 1: choice ~ countries + sanctions  
Model 2: choice ~ countries * sanctions  
Resid. Df Resid. Dev Df Deviance Pr(>Chi)  
1      6358      8674.9  
2      6354      8673.2  4    1.7263    0.7859
```